

Sai Spandana Chintapalli

Philadelphia, PA | schinta@engineering.upenn.edu | (929) 318-8382

[Website](#) | [Google Scholar](#) | [LinkedIn](#) | [ORCID](#)

RESEARCH PROFILE

Ph.D. researcher in biomedical machine learning and computational neuroimaging developing generative and representation-learning methods to characterize individualized and heterogeneous disease effects from population biomedical data. Research focuses on normative modeling, structured latent-variable learning, disease subtyping, counterfactual generation, and multimodal biomarkers of aging and neurodegeneration.

EDUCATION

- Ph.D. in Bioengineering**, University of Pennsylvania, Philadelphia, PA **Aug 2021–Present**
Expected completion: 2027. Advisors: Christos Davatzikos and Hao Huang; research mentor: Pratik Chaudhari.
- M.S. in Biomedical Engineering**, Columbia University, New York, NY **Aug 2018–Dec 2019**
GPA: 3.93/4.00; Shardashish Interschool Fellowship.
- B.Tech. in Biomedical Engineering**, Manipal Institute of Technology, Manipal, India **Jul 2014–Jul 2018**
GPA: 9.08/10.00; AICTE Scholarship, Academic Gold Medal, and merit awards.

FIRST- AND CO-FIRST-AUTHOR PUBLICATIONS

1. **Sai Spandana Chintapalli**, Pratik Chaudhari, and Christos Davatzikos. CASL-VAE: Learning Structured Latent Variables from Unpaired Data for Semi-supervised Clustering and Paired Sample Generation. *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. Accepted.
2. **Sai Spandana Chintapalli**, Pratik Chaudhari, and Christos Davatzikos. Unpaired-to-Paired Data Synthesis: Learning to Model Disease Effects via Contrastive Analysis. *CVPR Workshop on PHAROS-AIF-MIH*, 2026.
3. **Sai Spandana Chintapalli**, Sindhuja T. Govindarajan, Haochang Shou, Yong Fan, Hao Huang, and Christos Davatzikos. Parsing Disease Heterogeneity in Structural and Functional MRI-Derived Measures Using Normative Modeling and Generative Adversarial Networks. *Proceedings of SPIE Medical Imaging 13407*, 747–751, 2025. doi:10.1117/12.3040541.
4. **Sai Spandana Chintapalli**, Rongguang Wang, Zhijian Yang, Vasiliki Tassopoulou, et al. Generative Models of MRI-Derived Neuroimaging Features and Associated Dataset of 18,000 Samples. *Scientific Data* 11, 1330, 2024. doi:10.1038/s41597-024-04157-4. Co-first author.
5. T. E. Koker, **Sai Spandana Chintapalli**, S. Wang, et al. On Identification and Retrieval of Near-Duplicate Biological Images: A New Dataset and Protocol. *International Conference on Pattern Recognition*, 3114–3121, 2020. doi:10.1109/ICPR48806.2021.9412849. Co-first author.

SELECTED COLLABORATIVE PUBLICATIONS

1. Rongguang Wang, Vishnu Bashyam, Zhijian Yang, et al., including **Sai Spandana Chintapalli**. Applications of Generative Adversarial Networks in Neuroimaging and Clinical Neuroscience. *NeuroImage* 269, 119898, 2023. doi:10.1016/j.neuroimage.2023.119898.
2. Benjamin B. Sun, Stephanie J. Loomis, Fabrizio Pizzagalli, et al., including **Sai Spandana Chintapalli**. Genetic Map of Regional Sulcal Morphology in the Human Brain from UK Biobank Data. *Nature Communications* 13, 6071, 2022. doi:10.1038/s41467-022-33829-1.
3. Ziyi Huang, Yu Gan, Theresa Lye, et al., including **Sai Spandana Chintapalli**. Segmentation and Uncertainty Measures of Cardiac Substrates within Optical Coherence Tomography Images via Convolutional Neural Networks. *IEEE ISBI*, 1958–1961, 2020. doi:10.1109/ISBI45749.2020.9098495.

MANUSCRIPTS UNDER REVIEW

- Sai Spandana Chintapalli**, Sindhuja T. Govindarajan, Guray Erus, et al. Personalized Normative Modeling with RESTORE, a Deep Self-Supervised Framework, Reveals Individualized Disease- and Aging-Related Biological Changes in MRI and Proteomic Measures. Under review, 2026.

RESEARCH EXPERIENCE

- Ph.D. Researcher**, University of Pennsylvania, Philadelphia, PA **Aug 2021–Present**
Develop generative and representation-learning methods for individualized disease characterization from heterogeneous biomedical data, with applications to aging and neurodegeneration.
Developed CASL-VAE, a semi-supervised structured latent-variable framework for disentangling shared biological variability from target-specific variation and modeling disease subtypes and within-subtype heterogeneity from unpaired data.
Developed RESTORE, a self-supervised normative-modeling framework for estimating individualized reference states and subject-specific deviations without paired healthy–disease observations.
Evaluate methods across multi-cohort structural MRI, clinical, and CSF proteomic datasets using semi-synthetic benchmarks, independent biological validation, and longitudinal analyses.
- Data and Analytics Specialist IV**, Biogen Inc., Cambridge, MA **May 2020–Aug 2021**
Built MRI and PET acquisition and preprocessing workflows for clinical-study data, including metadata and pixel-data quality control, de-identification, and standardized organization on HPC systems.
Supported imaging biomarker research and UK Biobank imaging-phenotype harmonization analyses.
- Machine Learning Research Associate**, Harvard Medical School, Boston, MA **Jan 2020–May 2020**
Developed metric-learning and image-retrieval models for detecting near-duplicate biological images and co-created the BINDER benchmark dataset and protocol.
- R&D Intern**, Regeneron Pharmaceuticals, Tarrytown, NY **May 2019–Aug 2019**
Developed CNN-based segmentation and object-detection methods to quantify inflammatory cell clusters in retinal OCT images.
- Graduate Student Researcher**, Columbia University, New York, NY **Nov 2018–May 2019**
Supported deep-learning segmentation of cardiac OCT tissue for radiofrequency-ablation guidance.

INNOVATION & ENTREPRENEURSHIP

- Fovea / Tactile Vision**. Co-founded Fovea, a wearable sensory-substitution system mapping camera-derived spatial information to tactile signals for people with blindness.
- U.S. Patent 11,928,981 B2**, Tactile Vision, granted March 12, 2024; published as US 2023/0419859 A1. Inventors: Kevin Fan, Alaa Shamandy, and Spandana Chintapalli.

SELECTED PRESENTATIONS

- Individualized and Subtype-Aware Representation Learning for Biomedical Heterogeneity. Women in Machine Learning Workshop at ICML, 2026.
- Unpaired-to-Paired Data Synthesis: Learning to Model Disease Effects via Contrastive Analysis of Neuroimaging-Derived Features. PHAROS-AIF-MIH Workshop, CVPR, 2026.
- Two Sides of a Coin: Unraveling Disease Heterogeneity and Normal Biological Variability. UPenn AI in Medicine Symposium, 2026.
- Beyond Group Analysis: Deep Learning Methods for Personalized Diagnosis, Prognosis and Subtyping in Aging and Dementia. UPenn AI2D Symposium, 2025.
- Parsing Disease Heterogeneity: GAN-Based Normative Modeling for Individualized Disease Characterization. OHBM, 2024; updated presentation at OHBM, 2025.

AWARDS AND HONORS

- Women in Machine Learning Travel Award, 2026**. \$1,068 in travel support and ICML conference registration.
- Second Place Poster Award, AI2D Symposium, University of Pennsylvania, 2025**
- Winning Team, University of Toronto Entrepreneurship Hatchery NEST Demo Day, 2021**. Fovea was one of five winning teams selected from 17 finalists to share CAD 80,000 in seed funding.
- Grand Prize, MedHacks, Johns Hopkins University, 2019**. Best hack among 59 teams; \$2,500 award.
- Shardashish Interschool Fellowship, Columbia University, 2018–2019**. \$50,000 tuition support.
- AICTE Scholarship, Manipal Institute of Technology, 2014–2018**. 70% tuition waiver.
- Academic Gold Medals and Academic Excellence Awards, Manipal Institute of Technology, 2016–2018**.

LEADERSHIP & PROFESSIONAL SERVICE

Reviewer

Contributed to ICML and NeurIPS 2026 peer reviews as well as Nature Protocols paper review under faculty mentorship; the ICML reviewer account received **Gold Reviewer recognition (top 25%)**.

Diversity, Equity, and Inclusion Chair, Graduate Association of Bioengineers, University of Pennsylvania **2023–2024**

Represent graduate-student DEI priorities and organize inclusive professional-development programming, including a graduate career panel with invited speakers.

Board Member and Co-Publicity Head, Biomedical Engineering Society of India, Manipal Chapter **May 2016–Aug 2017**
Led event publicity and social media and launched the biannual magazine Biomed Chronicles.

Writer, MIT Post **Sep 2016–Jun 2017**
Reported on university events for Manipal Institute of Technology's official student media organization.

TEACHING EXPERIENCE

Course Assistant, Biomedical Imaging, Columbia University, New York, NY **Fall 2019**
Graded assignments, held office hours, and supported coding sessions.

Course Assistant, Digital Image Processing, Columbia University, New York, NY **Spring 2019**
Graded assignments and held office hours for undergraduate and graduate students.

TECHNICAL SKILLS

Programming: Python, R, MATLAB, SQL, shell scripting.

Machine Learning: PyTorch, TensorFlow/Keras, scikit-learn; generative modeling, variational inference, representation learning, Deep clustering, Computer Vision.

Biomedical Data: MRI, PET, OCT, multimodal clinical and omics data; DICOM, NIFTI.

Computing: HPC, Databricks, AWS, Git.